

## WEATHER FORECAST –WEB PAGE

### AIM:

o design and develop a **14-day weather forecast web application** using **HTML, CSS, and JavaScript** that fetches weather data from an API and displays the current weather along with daily forecast details such as temperature, weather condition, humidity, wind speed, and rainfall

### DESCRIPTION:

The Weather Forecast application allows the user to select a city and view its current weather information. JavaScript uses the **Open-Meteo Geocoding and Weather APIs** to obtain the location and weather data. The application dynamically displays the current temperature, humidity, wind speed, and weather condition, followed by **14 individual forecast cards** showing the date, weather condition, maximum and minimum temperatures, and rainfall. HTML provides the structure, CSS is used for styling and layout, and JavaScript handles API requests and dynamic data display.

### CODES:

#### HTML:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>14-Day Weather Forecast</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="container">

    <h1>☁ 14-Day Weather Forecast</h1>

    <!-- City Selection -->
    <div class="search-box">

      <select id="citySelect">
```

```

        <option value="">-- Select a City --</option>

        <option value="Hyderabad">Hyderabad</option>
        <option value="Bengaluru">Bengaluru</option>
        <option value="Chennai">Chennai</option>
        <option value="Mumbai">Mumbai</option>
        <option value="Delhi">Delhi</option>
        <option value="Kolkata">Kolkata</option>
        <option value="Pune">Pune</option>
        <option value="Vijayawada">Vijayawada</option>
        <option value="Visakhapatnam">Visakhapatnam</option>
        <option value="Tirupati">Tirupati</option>

    </select>

    <button id="searchBtn">Get Weather</button>

</div>

<p id="message"></p>

<!-- Weather Information -->
<div id="weather" class="hidden">

    <div class="current-weather">

        <h2 id="location"></h2>

        <div class="temperature">
            <span id="temperature"></span>°C
        </div>

        <p id="description"></p>

        <div class="details">

            <div>
                ☼
                <strong>Humidity</strong>
                <span id="humidity"></span>%
            </div>

            <div>
                ☼
                <strong>Wind</strong>
                <span id="wind"></span> km/h
            </div>

        </div>

    </div>

</div>

```

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```
<!-- 14-Day Forecast -->
<h2 class="forecast-title">
  14-Day Forecast
</h2>

<div id="forecast" class="forecast"></div>

</div>

</div>

<script src="script.js"></script>

</body>

</html>
```

CSS:

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: "Trebuchet MS", Arial, sans-serif;
  background: linear-gradient(135deg, #667eea, #764ba2);
  min-height: 100vh;
  padding: 30px;
}

.container {
  max-width: 1000px;
  margin: auto;
}

h1 {
  text-align: center;
  color: white;
  margin-bottom: 25px;
}

/* Search Area */

.search-box {
  display: flex;
  gap: 10px;
```

```
margin-bottom: 20px;
}

.search-box select {
  flex: 1;
  padding: 14px;
  border: none;
  border-radius: 8px;
  font-size: 16px;
  background: white;
  cursor: pointer;
}

.search-box button {
  padding: 14px 25px;
  border: none;
  border-radius: 8px;
  background: #22223b;
  color: white;
  cursor: pointer;
  font-size: 16px;
}

.search-box button:hover {
  background: #4a4e69;
}

#message {
  text-align: center;
  color: white;
  margin: 15px;
  font-size: 17px;
}

/* Current Weather */

.current-weather {
  background: rgba(255, 255, 255, 0.95);
  padding: 30px;
  border-radius: 15px;
  text-align: center;
  margin-bottom: 25px;
  box-shadow: 0 5px 15px rgba(0, 0, 0, 0.15);
}

#location {
  font-size: 28px;
  margin-bottom: 15px;
  color: #22223b;
}

.temperature {
  font-size: 60px;
```

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```
font-weight: bold;
color: #5a189a;
}

#description {
  font-size: 20px;
  margin: 10px 0 25px;
  color: #444;
}

.details {
  display: flex;
  justify-content: center;
  gap: 60px;
}

.details div {
  display: flex;
  flex-direction: column;
  gap: 5px;
  color: #333;
}

/* Forecast */

.forecast-title {
  color: white;
  text-align: center;
  margin-bottom: 15px;
}

.forecast {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 15px;
}

.forecast-card {
  background: white;
  padding: 18px;
  border-radius: 12px;
  text-align: center;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.12);
}

.forecast-card:hover {
  transform: translateY(-3px);
}
```

```
.forecast-card h3 {
  margin-bottom: 10px;
  color: #22223b;
}

.forecast-card p {
  margin: 8px 0;
  font-size: 14px;
}

.forecast-card .temp {
  font-size: 23px;
  font-weight: bold;
  color: #5a189a;
}

.forecast-card .min {
  color: #666;
}

.forecast-card .rain {
  color: #168aad;
}

.hidden {
  display: none;
}

/* Mobile */

@media (max-width: 800px) {

  .forecast {
    grid-template-columns: repeat(2, 1fr);
  }

}

@media (max-width: 500px) {

  .search-box {
    flex-direction: column;
  }

  .forecast {
    grid-template-columns: 1fr;
  }

  .details {
    gap: 25px;
  }

}
```

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```
}
```

JS:

```
// Get HTML Elements

const citySelect = document.getElementById("citySelect");
const searchBtn = document.getElementById("searchBtn");

const weatherDiv = document.getElementById("weather");
const message = document.getElementById("message");

const locationElement = document.getElementById("location");
const temperatureElement = document.getElementById("temperature");
const descriptionElement = document.getElementById("description");

const humidityElement = document.getElementById("humidity");
const windElement = document.getElementById("wind");

const forecastDiv = document.getElementById("forecast");

// Button Event

searchBtn.addEventListener("click", getWeather);

// Get Weather Function

async function getWeather() {

    const city = citySelect.value;

    // Check city selection

    if (city === "") {

        message.textContent = "Please select a city.";
        weatherDiv.classList.add("hidden");

        return;
    }

    message.textContent = "Loading weather data...";
    weatherDiv.classList.add("hidden");

    try {
```

```
// Geocoding API
// City Name → Latitude + Longitude

const geoURL =
  `https://geocoding-api.open-
meteo.com/v1/search?name=${encodeURIComponent(city)}&count=1&language=en&format=json`;

const geoResponse = await fetch(geoURL);

if (!geoResponse.ok) {
  throw new Error("Unable to find city.");
}

const geoData = await geoResponse.json();

if (!geoData.results ||
  geoData.results.length === 0) {

  throw new Error("City not found.");
}

// Get Location

const location = geoData.results[0];

const latitude = location.latitude;
const longitude = location.longitude;

// Weather API

const weatherURL =
  `https://api.open-meteo.com/v1/forecast` +
  `?latitude=${latitude}` +
  `&longitude=${longitude}` +
  `&current=temperature_2m,relative_humidity_2m,wind_speed_10m,weather_code` +
  `&daily=weather_code,temperature_2m_max,temperature_2m_min,precipitation_sum` +
  `&timezone=auto` +
  `&forecast_days=14`;

// Send Request

const weatherResponse = await fetch(weatherURL);

if (!weatherResponse.ok) {
  throw new Error("Unable to fetch weather data.");
}

const weatherData = await weatherResponse.json();
```



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```
// Display Weather

displayWeather(location, weatherData);

message.textContent = "";
weatherDiv.classList.remove("hidden");

}

catch (error) {

    message.textContent = error.message;
    weatherDiv.classList.add("hidden");

}
}

// Display Current Weather

function displayWeather(location, data) {

    locationElement.textContent =
        `${location.name}, ${location.country}`;

    temperatureElement.textContent =
        Math.round(data.current.temperature_2m);

    descriptionElement.textContent =
        getWeatherDescription(
            data.current.weather_code
        );

    humidityElement.textContent =
        data.current.relative_humidity_2m;

    windElement.textContent =
        data.current.wind_speed_10m;

    // Display 14-Day Forecast

    displayForecast(data.daily);
}
```

```
// Display Forecast Cards

function displayForecast(daily) {

    // Remove previous cards

    forecastDiv.innerHTML = "";

    // Loop through 14 days

    for (let i = 0; i < daily.time.length; i++) {

        // Convert date

        const date = new Date(daily.time[i]);

        // Get day name

        const dayName =
            date.toLocaleDateString(
                "en-US",
                {
                    weekday: "short"
                }
            );

        // Get date

        const dateText =
            date.toLocaleDateString(
                "en-US",
                {
                    day: "numeric",
                    month: "short"
                }
            );

        // Create Card

        const card =
            document.createElement("div");

        card.classList.add("forecast-card");

        // Add Information

        card.innerHTML = `
```

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```
<h3>${dayName}</h3>

<p>${dateText}</p>

<p>
    ${getWeatherDescription(
        daily.weather_code[i]
    )}
</p>

<p class="temp">
    ${Math.round(
        daily.temperature_2m_max[i]
    )}°C
</p>

<p class="min">
    Min:
    ${Math.round(
        daily.temperature_2m_min[i]
    )}°C
</p>

<p class="rain">
    ☐ Rain:
    ${daily.precipitation_sum[i]} mm
</p>

`;

// Add Card

forecastDiv.appendChild(card);

}
}

// Weather Code Description

function getWeatherDescription(code) {

    const weatherCodes = {

        0: "Clear sky",

        1: "Mainly clear",
```

```
2: "Partly cloudy",

3: "Overcast",

45: "Fog",

48: "Fog",

51: "Light drizzle",

53: "Moderate drizzle",

55: "Dense drizzle",

61: "Slight rain",

63: "Moderate rain",

65: "Heavy rain",

71: "Slight snowfall",

73: "Moderate snowfall",

75: "Heavy snowfall",

80: "Slight rain showers",

81: "Moderate rain showers",

82: "Heavy rain showers",

95: "Thunderstorm"

};

return weatherCodes[code] || "Unknown weather";
}
```

**OUTPUT:**

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